LoanBazaar®
Secure Distribution Venture Pvt. Ltd.

What Is Infrastructure Funding ?

Developed by www.loanbazaaronline.com

TABLE OF CONTENT

Sr.No.	TITLE	Page No.
1.	**Module 1:Product Overview & Macroeconomic Paradigm of Infrastructure Funding**	**3-6**
2.	**Module 2:Structural Architecture & Financing Modalities of Infrastructure Funding**	**7-10**
3.	**Module 3: Comparative Capital Architecture Matrix**	**11-13**
4.	**Module 4: Common Eligibility Criteria, Credit Appraisal & Underwriting Framework**	**14-16**
5.	**Module 5:Security Architecture, Escrow TRA Waterfalls & Legal Covenants**	**17-19**
6.	**Module 6:Indicative Tenure, Pricing Mechanics, Accounting & Tax Shields**	**20-22**
7.	**Module 7:Suitable Industries, Business Profiles & Sub-Sector Applications**	**23-24**
8.	**Module 8: Key Advantages and Strategic Limitations of Infrastructure Funding**	**25**
9.	**Module 9: Comprehensive Real-World Case Study (Indian Rupee Models)**	**26-29**
10.	**Module 10:Regulatory Compliance, RBI Guidelines (2025/2026) & DCCO Framework**	**30**
11.	**Module 11: Professional Certification Examination & Detailed Rationales**	**31-38**

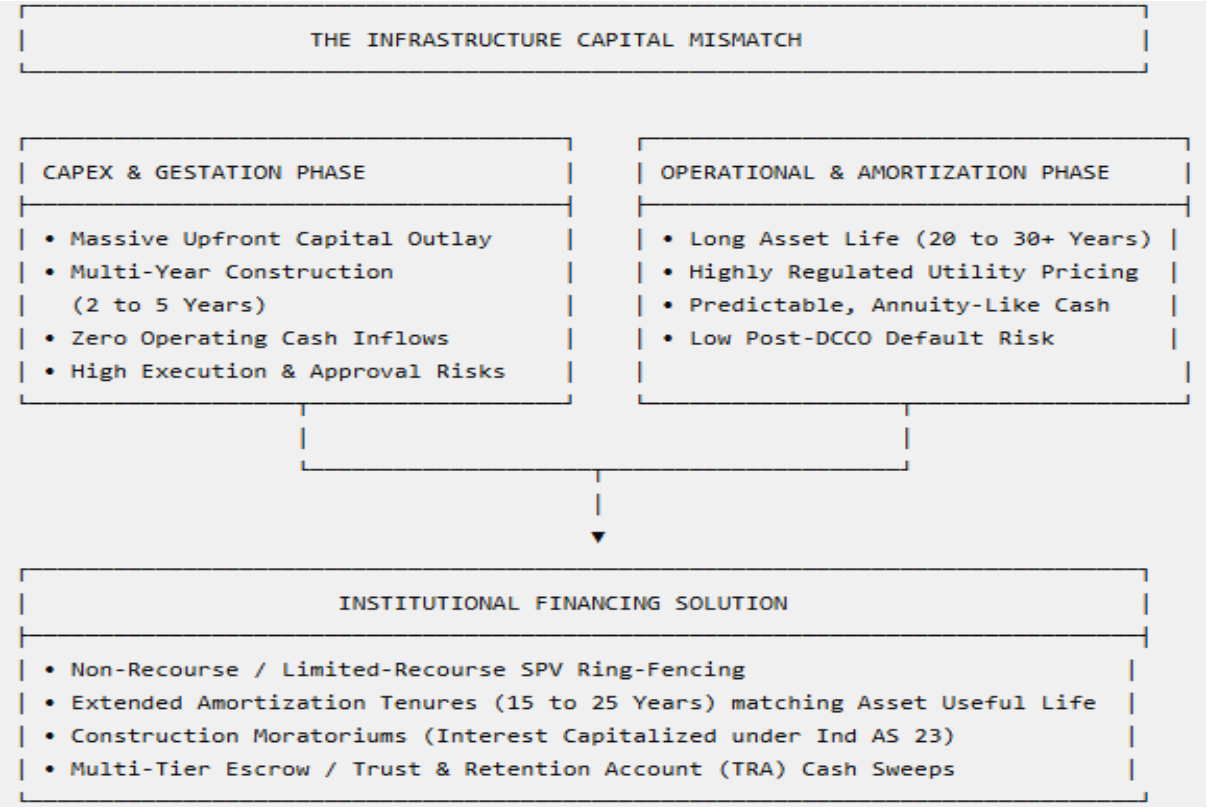
Copyright Notice: © 2026 www.loanbazaronline.com. All rights reserved.

Module 1: Product Overview & Macroeconomic Paradigm of Infrastructure Funding

Infrastructure Funding represents the specialized institutional credit, capital structuring, and debt syndication framework designed to mobilize long-term capital for large-scale physical, social, and digital infrastructure assets across India. Unlike standard corporate loans that rely on the historical balance sheet and multi-product cash flows of an established operating company, infrastructure funding is predominantly structured around the project's standalone economic viability, projected cash flows, contractual agreements, and asset-backed security stack.

1.1 The Fundamental Infrastructure Capital Mismatch Paradox

The financing of nation-building infrastructure assets—such as national expressways, high-voltage power transmission grids, solar and wind energy parks, deep-water ports, greenfield airports, urban metro rail networks, data centers, and bulk water treatment systems—presents a severe structural challenge known in corporate finance as the **Infrastructure Capital Mismatch Paradox**.



Copyright Notice: © 2026 www.loanbazaronline.com. All rights reserved.

Infrastructure assets require massive capital investments during their initial development phase, often running into hundreds or thousands of Crores of Rupees ($ ext{₹}$). During this multi-year construction window, the project generates zero revenue while incurring heavy interest costs, engineering outlays, and statutory compliance expenses. However, once the asset reaches its **Date of Commercial Operation (DCCO)**, it transitions into a highly stable, utility-like economic engine characterized by predictable, long-term, annuity-like cash flows spanning 20 to 30+ years.

If a corporate sponsor attempts to fund a $ ext{₹1,000 Crore}$ highway or power plant using traditional 3-to-5 year commercial bank term loans, the annual debt service obligation during the early operational years will immediately swamp project revenue, triggering rapid cash insolvency and default—even if the underlying asset is fundamentally sound and profitable over a 20-year horizon. Infrastructure Funding resolves this paradox by aligning the repayment structure, interest rate benchmarks, grace periods, and collateral covenants directly with the 20-to-30 year cash-generation lifecycle of the physical asset.

1.2 Evolution of Infrastructure Financing in India

The landscape of infrastructure financing in India has undergone a radical transformation over the past three decades, evolving through distinct regulatory and structural epochs:

1. **The Public Sector Banking & Budgetary Epoch (Pre-2008):** Infrastructure development was primarily funded directly through state/central government budgetary allocations or through heavy balance-sheet lending by Indian Public Sector Banks (PSBs). This era suffered from severe asset-liability mismatches (ALM) on bank balance sheets, as banks funded 15-year infrastructure assets using 3-to-5 year fixed-term retail deposits, leading to high Non-Performing Assets (NPAs) when projects faced land acquisition and environmental delays.
2. **The Public-Private Partnership (PPP) & Concession Epoch (2008–2018):** To crowd in private capital, the Government of India and state agencies introduced structured PPP models, including **Build-Operate-Transfer (BOT) Toll**, **BOT Annuity**, and **Design-Build-Finance-Operate-Transfer (DBFOT)** frameworks. Regulatory bodies such as the National Highways Authority of India (NHAI), Solar Energy Corporation of India (SECI), and State Nodal Agencies established standardized long-term Concession Agreements and Power Purchase Agreements (PPAs).
3. **The Hybrid Annuity Model (HAM) & DFI Epoch (2018–2024):** To resolve private sector risk aversion caused by traffic and demand volatility, NHAI pioneered the **Hybrid Annuity Model (HAM)**, where the government funds 40% of the project cost in cash tranches

Copyright Notice: © 2026 www.loanbazaronline.com. All rights reserved.

during construction, leaving the private sponsor to arrange only 60% through debt and equity. Concurrently, specialized Development Finance Institutions (DFIs) such as the **National Bank for Financing Infrastructure and Development (NaBFID)**, **India Infrastructure Finance Company Limited (IIFCL)**, **Power Finance Corporation (PFC)**, **REC Limited**, and **IREDA** were empowered to provide long-tenure senior debt.

4. **The Yield Monetization & Asset Recycling Epoch (2024–2026+):** The modern era is defined by secondary market liquidity, asset monetization via **Infrastructure Investment Trusts (InvITs)**, cross-border External Commercial Borrowings (ECBs), green bonds, and the **National Monetisation Pipeline (NMP)**. Fully operational, revenue-generating infrastructure assets are routinely refinanced or securitized into InvITs, freeing up primary banking capital to fund new greenfield projects.

1.3 Statutory, Regulatory & Policy Framework Governing Indian Infrastructure Funding

Infrastructure Funding in India operates under a highly sophisticated, multi-layered regulatory architecture designed to enforce credit discipline, protect bank depositors, and ensure time-bound project execution:

- **RBI Harmonized Master List of Infrastructure Sub-Sectors:** Maintained by the Reserve Bank of India (RBI) and Ministry of Finance, this master list defines eligible infrastructure sub-sectors entitled to long-term concessional debt, higher bank exposure limits, flexible restructuring, and external commercial borrowing access. Sub-sectors are broadly classified into Transport & Logistics, Energy, Water & Sanitation, Communication, and Social & Commercial Infrastructure.
- **RBI Guidelines on Project Financing (2024–2026 Framework):** Enforces strict credit risk management rules for commercial banks and NBFCs financing infrastructure projects. The framework mandates calibrated standard asset provisioning during the construction phase (up to 5.0% during construction, stepping down to 2.5% upon COD and 1.0% after two years of operational stability), strict DCCO tracking, and mandatory Lender's Independent Engineer (LIE) audits.
- **RBI Prudential Norms on Income Recognition, Asset Classification and Provisioning (IRACP):** Defines default timelines, Special Mention Account (SMA-0, SMA-1, SMA-2) triggers, and 90-day Non-Performing Asset (NPA) classification rules. It provides specific guidelines for permissible DCCO extensions without classifying the account as an NPA if delays are caused by legal disputes, court injunctions, or statutory approval delays beyond promoter control.

Copyright Notice: © 2026 www.loanbazaronline.com. All rights reserved.

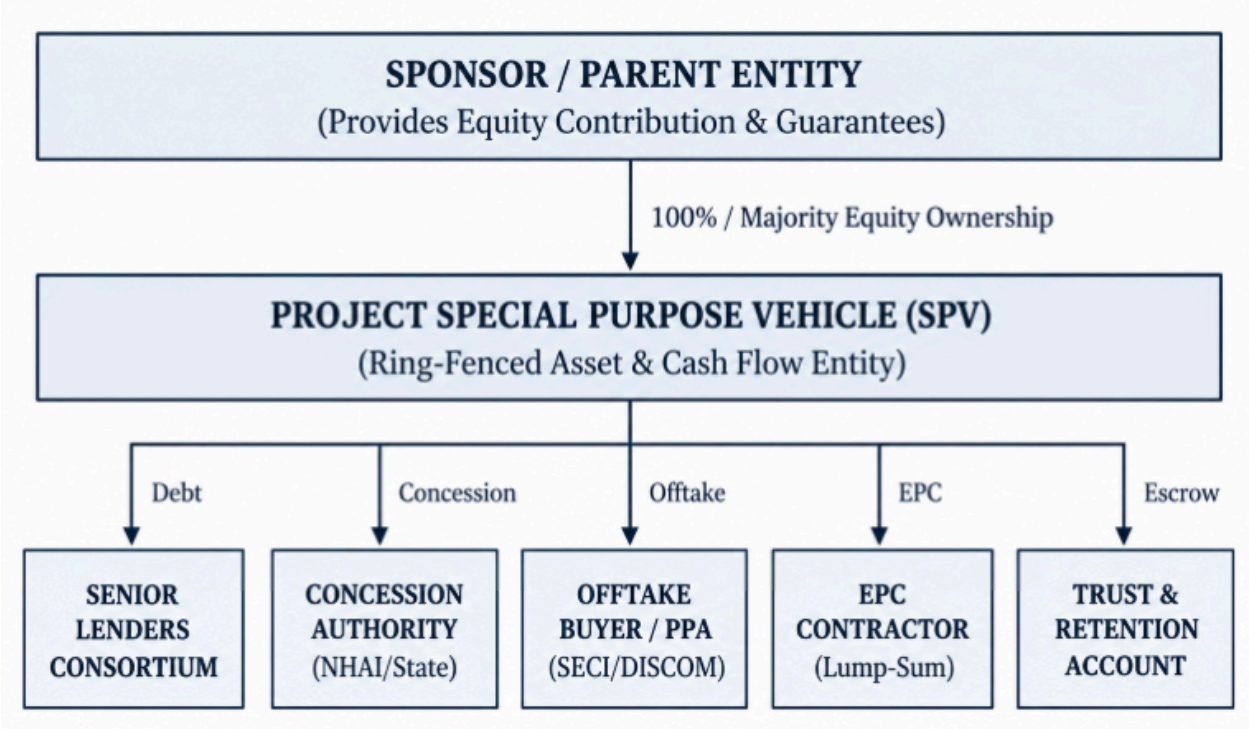
- **The Insolvency and Bankruptcy Code, 2016 (IBC):** Governs debt enforcement and resolution for distressed infrastructure SPVs. Under Section 5(8) of the IBC, infrastructure lenders qualify as Financial Creditors holding voting rights in the Committee of Creditors (CoC) to execute time-bound corporate insolvency resolution processes (CIRP).
- **Statutory Charge Registration & Escrow Frameworks:** Enforces mandatory registration of all asset mortgages, hypothecations, and share pledges on the **Central Registry of Securitisation Asset Reconstruction and Security Interest of India (CERSAI)** portal and **Registrar of Companies (ROC) via Form CHG-1** within 30 days of creation under Section 77 of the Companies Act, 2013.

Copyright Notice: © 2026 www.loanbazaronline.com. All rights reserved.

Module 2: Structural Architecture & Financing Modalities of Infrastructure Funding

2.1 The Non-Recourse Special Purpose Vehicle (SPV) Architecture

The foundational pillar of modern Infrastructure Funding is the **Special Purpose Vehicle (SPV) Architecture**. Rather than raising debt on its own corporate balance sheet, the parent sponsor company incorporates an independent, ring-fenced subsidiary company (the SPV) dedicated exclusively to constructing, owning, operating, and servicing the specific infrastructure asset.



Key Elements of the SPV Structure:

1. **Ring-Fenced Bankruptcy Remote Status:** The assets, liabilities, cash inflows, and contractual rights of the SPV are legally segregated from the parent sponsor. If the parent company faces financial distress or bankruptcy, the infrastructure project SPV remains completely insulated, protecting project operations and lender cash flows. Conversely, if the SPV defaults, senior lenders have recourse primarily to the SPV's assets and escrow cash flows, rather than the parent sponsor's balance sheet (limited-recourse or non-recourse financing).
2. **Dedicated Single-Purpose Charter:** The Memorandum of Association (MOA) and Articles of Association (AOA) of the SPV strictly restrict its business activities exclusively to executing and operating the specific infrastructure project. It is legally prohibited from

Copyright Notice: © 2026 www.loanbazaronline.com. All rights reserved.

undertaking unrelated commercial activities, acquiring outside equity, or issuing third-party guarantees.

3. **Controlled Cash Sweep Architecture:** All project revenues—whether toll collections, government annuities, PPA electricity tariffs, airport landing fees, or data center lease rentals—are legally mandated to flow directly into a single, bank-controlled **Trust and Retention Account (TRA)** Escrow System, preventing any unauthorized diversion of funds by the promoter.

2.2 Core Financing Modalities in Indian Infrastructure Funding

Depending on the underlying sub-sector, revenue model, and risk allocation, Infrastructure Funding in India is structured across six primary capital modalities:

1. Hybrid Annuity Model (HAM) Project Financing

Pioneered primarily in the highways and urban water treatment sectors, HAM combines public grant funding with private debt and equity: * **Government Construction Grant (40%):** Paid out in 5 equal tranches of 8% each by the Concession Authority (e.g., NHAI) linked to physical construction progress milestones (20%, 40%, 60%, 80%, 100%). * **Private Sponsor Contribution (60%):** Sourced through a combination of Equity (typically 12% to 15% of total project cost) and Senior SPV Debt (45% to 48% of total project cost). * **Annuity Debt Servicing:** Post-COD, the Concession Authority pays the SPV bi-annual inflation-indexed annuities over 15 years, covering: - Fixed semi-annual principal reimbursement tranches. - Interest on the residual benchmarked balance (indexed to the prevailing RBI Bank Rate + spread). - Inflation-adjusted Operation & Maintenance (O&M) allowances. * **Risk Profile:** Zero traffic/volume risk; credit risk is transferred to the sovereign/statutory Concession Authority.

2. Build-Operate-Transfer (BOT) Toll & User-Fee Debt Structures

Under BOT Toll models, the SPV constructs the infrastructure asset and earns revenue by directly collecting user fees/tolls from commercial traffic over a 20-to-30 year concession period: * **Sponsor & Lender Exposure:** High exposure to construction completion risks, traffic growth rates, GDP elasticity, and alternate route diversion risks. * **Financial Covenants:** Requires higher debt service coverage ratios ($ ext{DSCR} 1.45x - 1.60x$), robust traffic sensitivity stress-testing, and dynamic cash sweep mechanisms that accelerate debt amortization during high-traffic quarters.

3. Power Purchase Agreement (PPA) & Tariff-Based Offtake Debt

Utilized in renewable energy (solar, wind, hybrid), thermal power, and transmission grids: * **Contractual Backing:** Governed by long-term 25-year PPAs executed with creditworthy central

Copyright Notice: © 2026 www.loanbazaronline.com. All rights reserved.

counterparties (e.g., Solar Energy Corporation of India - SECI, NTPC) or state electricity distribution companies (DISCOMs). * **Tariff Structure:** Fixed single-part tariff ($ ext{₹/kWh}$) or two-part tariff (Capacity Charge for availability + Energy Charge for fuel pass-through). * **Financing Mechanics:** Debt is sized based on guaranteed P-50 / P-90 plant load factors (PLF), minimum generation guarantees, and statutory Payment Security Mechanisms (PSMs) such as Letters of Credit and escrow accounts backed by state/central guarantees.

4. Take-Out Financing & Post-DCCO Debt Refinancing

A critical liquidity mechanism where high-cost, short-tenure construction debt provided by commercial banks during the high-risk construction phase is replaced ("taken out") post-DCCO by long-term institutional investors: * **Refinancing Entities:** Specialized DFIs (NaBFID, IIFCL), Insurance Companies (LIC), Pension Funds, Infrastructure Debt Funds (IDFC/NBFC-IDFs), and InvITs. * **Benefits:** Slashes borrowing interest rates by 150 to 300 basis points post-COD, extends debt amortization out to 15–20 years, and frees up primary bank capital limits to fund new greenfield projects.

5. Infrastructure Investment Trusts (InvITs)

Regulated under SEBI (Infrastructure Investment Trusts) Regulations, 2014, InvITs act as collective investment vehicles that pool operational, revenue-generating infrastructure assets: * **Capital Structure:** InvITs raise equity unit capital from institutional/retail investors and issue low-cost senior NCDs/bonds to refinance SPV-level bank debt. * **Distribution Mandate:** Mandatory distribution of at least 90% of Net Distributable Cash Flows (NDCF) to unit holders semi-annually, providing predictable dividend/interest yields.

6. Multilateral & Cross-Border External Commercial Borrowings (ECBs)

High-value infrastructure projects routinely tap international capital markets for long-term foreign currency loans or Rupee-denominated Masala Bonds: * **Multilateral DFIs:** Asian Development Bank (ADB), Asian Infrastructure Investment Bank (AIIB), New Development Bank (NDB), and International Finance Corporation (IFC). * **RBI ECB Framework:** Governed by RBI Master Direction No. 5, allowing infrastructure SPVs to raise up to $ ext{USD 750 Million}$ per financial year under the Automatic Route with a Minimum Average Maturity Period (MAMP) of 3 to 5 years (and up to 10 years for specific capital uses), subject to RBI All-In-Cost ceilings ($ ext{SOFR} + 450 ext{ bps}$).

Copyright Notice: © 2026 www.loanbazaronline.com. All rights reserved.

2.3 The Web of Interlocking Commercial Contracts

In non-recourse infrastructure funding, cash flow security is enforced through an interlocking ecosystem of standardized commercial contracts:

WEB OF INTERLOCKING CONTRACTS		
1.	CONCESSION AGREEMENT / PPA	Grants right to build, operate & collect revenue
2.	LUMP-SUM TURNKEY EPC CONTRACT	Guarantees fixed cost, time & performance LDs
3.	O&M AGREEMENT	Mandates long-term operating efficiency & uptime
4.	FUEL / INPUT SUPPLY (FSA)	Secures raw material volume, quality & pricing
5.	ESCROW TRA AGREEMENT	Directs all revenue into bank-controlled accounts
6.	SUBSTITUTION RIGHTS AGREEMENT	Allows lenders to replace promoter on default

Copyright Notice: © 2026 www.loanbazaronline.com. All rights reserved.

Module 3: Comparative Capital Architecture Matrix

The following high-density comparative matrix evaluates the structural, economic, and regulatory differences across the primary financing modalities utilized in Indian Infrastructure Funding:

Parameter / Feature	Non-Recourse SPV Infrastructure Loan	Corporate Balance Sheet Loan	Hybrid Annuity Model (HAM) Project Debt	InvIT Refinancing Debt (Post-COD)	Subordinated / Mezzanine AIF Debt
Primary Borrower Entity	Independent Ring-Fenced SPV Ltd.	Operating Parent / Corporate Sponsor	Greenfield Highway Project SPV	SEBI-Registered InvIT Trust / SPV	Project SPV / HoldCo
Parent Recourse Level	Non-Recourse / Limited to Sponsor Equity Commitments	Full Direct Recourse to Parent Balance Sheet	Limited Recourse during Construction; Non-Recourse Post-COD	Non-Recourse to Sponsor; Direct Recourse to InvIT Cash Flows	Subordinated Recourse; Equity Kicker / High Yield
Max Leverage (Debt:Equity)	$70:30 ext{ to } 80:$	$60:40 ex$	$75:25 ext{ (SPV Debt } pprox 45\%, ext{ Grant } 40\%, ext{ Equity } 15\%)$	Up to $70\% ext{ LTV}$	$80:20 ext{ to}$

Copyright Notice: © 2026 www.loanbazaronline.com. All rights reserved.

Parameter / Feature	Non-Recourse SPV Infrastructure Loan	Corporate Balance Sheet Loan	Hybrid Annuity Model (HAM) Project Debt	InvIT Refinancing Debt (Post-COD)	Subordinated / Mezzanine AIF Debt
Pricing Benchmark & Spread	EBLR / MCLR $+ 2.00\%extto$	MCLR / EBLR $+ 1.50\%$	RBI Bank Rate $+ 1.25\%ex$	NCD Yield / AAA Bond Rate $+ 0.75\%ex$	Fixed Coupon $14.00\%ext$ (Gross IRR)
Construction Moratorium	$2extto4extYec$ (Equal to Construction + 6 Months)	$6extto1$	$2extto3ext$ (Up to DCCO + First Annuity Date)	Zero (Asset is 100% Operational)	$2extto3ext$ (Interest Capitalized / PIK)
Repayment Amortization	$12extto18extY$ Post-COD (Door-to-Door)	$5extto7$ Amortization	$12extto15e$ Semi-Annual Sculpted Annuities	$10extto20e$ Bond Bullet / Sculpted Amortization	$3extto5ext$ Bullet / Refinancing Sweep
Primary Collateral Stack	First Mortgage on SPV Assets + Escrow TRA Charge + Share Pledge	General Fixed Asset Charge + Corporate Guarantee	First Charge on Annuity Escrow TRA + Mortgage on Concession Rights	Exclusive Charge on Pool Asset Cash Flows + Unit Pledge	Subordinated Charge on SPV Cash Flows / HoldCo Shares
Target Min DSCR Threshold	$\geq 1.35xext(A$	$\geq 1.50x$	$\geq 1.20xexi$	$\geq 1.40xexi$	$\geq 1.10xexi$

Copyright Notice: © 2026 www.loanbazaronline.com. All rights reserved.

Parameter / Feature	Non-Recourse SPV Infrastructure Loan	Corporate Balance Sheet Loan	Hybrid Annuity Model (HAM) Project Debt	InvIT Refinancing Debt (Post-COD)	Subordinated / Mezzanine AIF Debt
Escrow TRA Control Level	Mandatory 7-Tier Strict Waterfall Escrow	Standard Operating Account / Optional Escrow	Mandatory 6-Tier Semi-Annual Annuity Escrow TRA	Mandatory SEBI NDCF Distribution Escrow Account	Subordinated TRA Cash Sweep Account
Tax Shield & Deductibility	Sec 36(1)(iii) Full Interest Deductibility; Sec 80-IA	Sec 36(1)(iii) Standard Interest Deductibility	Sec 36(1)(iii) Full Interest Deductibility	Sec 10(23FC) / Sec 115UA Tax Exemption for Pass-Through	Sec 36(1)(iii) Deductibility subject to Thin Cap rules

Copyright Notice: © 2026 www.loanbazaronline.com. All rights reserved.

Module 4: Common Eligibility Criteria, Credit Appraisal & Underwriting Framework

The credit appraisal of an Infrastructure Funding proposal differs fundamentally from standard commercial lending. Because senior debt is serviced primarily from the future cash flows of a single, standalone project SPV, institutional underwriters perform rigorous, multi-dimensional due diligence spanning technical, commercial, legal, environmental, and financial domains.

4.1 Sponsor & Project Eligibility Covenants

1. Sponsor / Promoter Eligibility Benchmarks:

- **Demonstrated Technical Execution Track Record:** The parent promoter or lead consortium member must possess a proven history of successfully commissioning and operating projects of similar scale and complexity (e.g., executing at least 500 MW of solar capacity or 500 lane-kilometers of national highways).
- **Consortium Net Worth Covenants:** The sponsor group must demonstrate a minimum Tangible Net Worth (TNW) equal to at least 100% to 150% of the total promoter equity commitment required for the project SPV, ensuring ability to meet equity call notices and cost overrun undertakings.
- **Credit Rating & CIBIL Bureau Metrics:** Sponsor entities must maintain a minimum corporate credit rating of **CRISIL / ICRA / CARE BBB+ or higher**, alongside a **CIBIL Commercial Rank (CMR) of CMR 1 to CMR 4**. Promoters listed as Wilful Defaulters or Non-Cooperative Borrowers on RBI/CIC databases are strictly ineligible.

2. Project & Statutory Clearances Eligibility:

- **Fully Executed Offtake / Concession Contract:** Execution of a binding Concession Agreement with a government authority (NHAI, MoRTH, State PWD) or a long-term PPA with an creditworthy buyer (SECI, NTPC, Tier-1 DISCOM).
- **Land Availability Thresholds:**
  - *Highway/Road Projects:* Demonstration of unencumbered physical possession of at least **80% to 90% of the required Right of Way (RoW)** prior to financial closure disbursal, in compliance with NHAI and RBI guidelines.
  - *Solar/Wind/Thermal Projects:* 100% clear title or registered long-term leasehold rights (25 to 30 years) over project land.
- **Statutory Approval Clearances:** Receipt of final Environment Impact Assessment (EIA) clearance from the Ministry of Environment, Forest and Climate Change (MoEFCC), Forest Clearance (Stage-I & Stage-II), Central Electricity Authority (CEA) concurrence,

Copyright Notice: © 2026 www.loanbazaronline.com. All rights reserved.

Defense/Aviation NOCs, and Grid Connectivity Approval from POWERGRID / State Transmission Utility (STU).

4.2 Financial Underwriting Ratios, Formulas & Stress Testing

Lenders evaluate the debt capacity of an infrastructure SPV using a suite of specialized financial metrics, calculating exact coverage ratios under base-case and downside-stressed operating scenarios:

INFRASTRUCTURE FINANCIAL UNDERWRITING		
1.	**CFADS**	CFADS = EBITDA + Interest Income - Taxes Paid - Net NWC - Maintenance Capex
2.	**DSCR**	DSCR = CFADS / Total Annual Debt Service (Principal Amortization + Interest)
3.	**LLCR**	LLCR = NPV of CFADS over Loan Amortization Life / Total Debt Outstanding
4.	**PLCR**	PLCR = NPV of CFADS over Total Concession Asset Life / Total Debt Outstanding

1. Cash Flow Available for Debt Service (CFADS)

The primary cash metric utilized in infrastructure underwriting, CFADS reflects the actual net cash generated by the project SPV available to meet debt obligations:

$$extCFADS = extEBITDA + extInterestIncomeonEscrowDeposits - extCashCorporateTaxes - \Delta extNetWorkingCapital - extMandatoryMaintenanceCapex$$

2. Debt Service Coverage Ratio (DSCR)

Measures the annual cash buffer available to service senior debt (Principal Amortization + Pre-Tax Interest):

$$extDSCR_t = racextCFADS_t extPrincipalAmortization_t + extPre - TaxInterestCost_t$$

- **Target Benchmarks:**
  - *Annuity / HAM / SECI PPA Projects:* Minimum Average $ ext{DSCR} 1.25x - 1.30x$; Minimum Year $ ext{DSCR} 1.15x$.
  - *Merchant / BOT Toll Projects:* Minimum Average $ ext{DSCR} 1.45x - 1.60x$; Minimum Year $ ext{DSCR} 1.25x$.

Copyright Notice: © 2026 www.loanbazaronline.com. All rights reserved.

3. Loan Life Coverage Ratio (LLCR)

Evaluates debt coverage over the entire remaining tenor of the senior loan facility by discounting future CFADS at the weighted average cost of debt (K_d):

$$extLLCR_t = rac \sum_{k=t}^{T_{extloan}} racextCFADS_k\left(1 + K_d\right)^{k-t} extOutstandingSeniorDebtBalance_t$$

* **Underwriting Covenant:** Target $ ext{LLCR} 1.35x - 1.50x$, ensuring that even if an individual year experiences temporary cash tightness, the total NPV of project cash flows comfortably covers the debt principal.

4. Project Life Coverage Ratio (PLCR)

Evaluates debt coverage over the entire remaining operational/concession life of the physical asset ($T_{extconcession}$), including the “tail period” after the loan matures:

$$extPLCR_t = rac \sum_{k=t}^{T_{extconcession}} racextCFADS_k\left(1 + K_d\right)^{k-t} extOutstandingSeniorDebtBalance_t$$

* **Underwriting Covenant:** Target $ ext{PLCR} 1.60x - 1.80x$. The difference between LLCR and PLCR represents the “Tail Period Equity Cushion” available for debt refinancing or extension.

5. Stress Testing & Sensitivity Sensitivity Covenants

Before issuing a sanction letter, lenders simulate severe downside sensitivity models: * **Cost Overrun Sensitivity:** Simulating a 10% to 20% spike in EPC construction costs. * **Interest Rate Shock:** Stressing the floating benchmark rate by $+ 150extto + 250extbps$. * **Traffic / Generation Drop:** Simulating a 15% reduction in toll traffic volume or a drop in solar/wind generation from P-50 down to P-90 levels. * **Payment Delays:** Testing a 90-to-180 day delay in DISCOM tariff or government annuity remittances.

Copyright Notice: © 2026 www.loanbazaronline.com. All rights reserved.

Module 5: Security Architecture, Escrow TRA Waterfalls & Legal Covenants

To enforce non-recourse credit ring-fencing, Infrastructure Funding utilizes a comprehensive, legally tight security package combined with a bank-controlled payment escrow system.

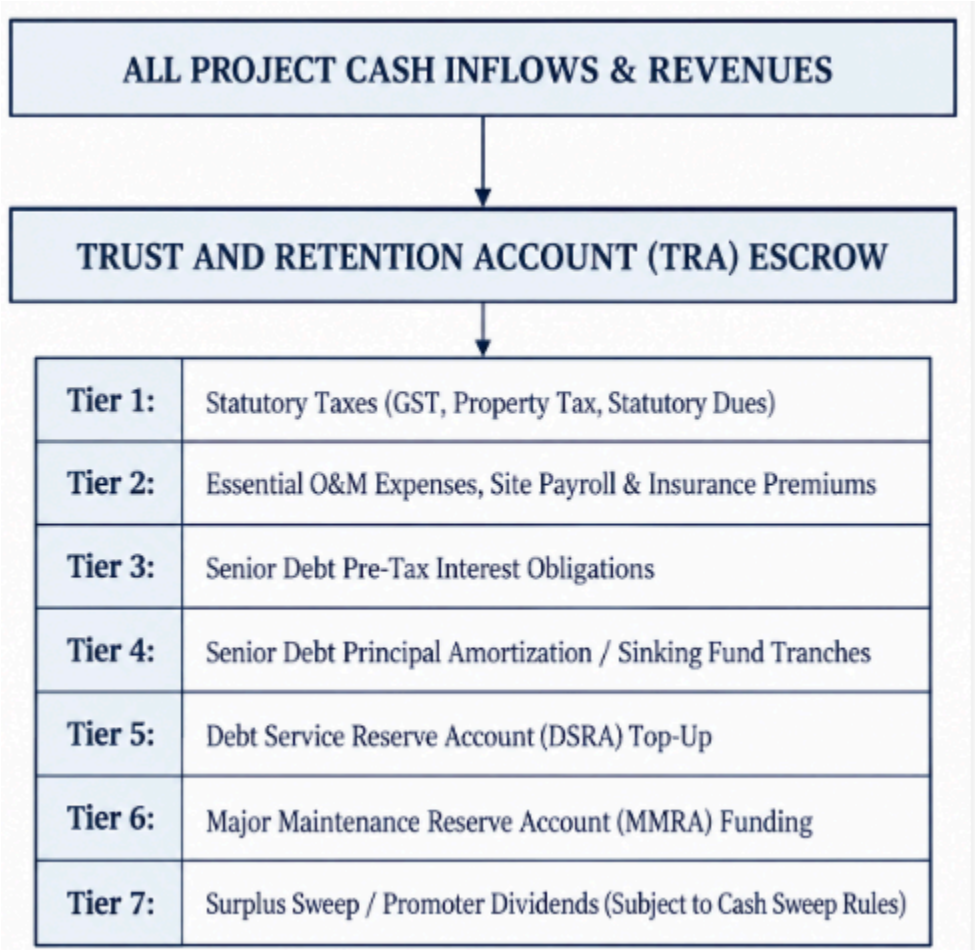
5.1 The Comprehensive Security Package

1. **Section 58(f) Equitable Mortgage / Registered Mortgage:** First ranking pari-passu mortgage over all immovable properties, land, rights-of-way, civil structures, buildings, and leasehold rights of the project SPV under the Transfer of Property Act, 1882.
2. **Hypothecation of Movable Assets & Receivables:** First ranking charge over all plant, machinery, turbines, equipment, raw material stock, inventory, work-in-progress, unbilled revenues, and trade receivables under Section 2(1)(n) of the SARFAESI Act, 2002.
3. **Pledge of Equity Shares:** Irrevocable pledge of at least **51% of the total voting equity share capital** (along with preference shares and CCPS) of the project SPV held by the promoter, preventing any unauthorized change of management or corporate control without prior lender approval.
4. **Assignment of Project Contracts & Intangibles:** Absolute legal assignment by way of security over all project agreements, including the Concession Agreement, PPA, EPC Lump-Sum Contract, O&M Agreement, Fuel Supply Agreements, Performance Bank Guarantees, Insurance Policies, and Government Grants/Subsidies.
5. **Debt Service Reserve Account (DSRA):** A dedicated, funded cash reserve or bank guarantee equal to **1 to 2 quarters of principal and interest servicing obligations**, held in a blocked escrow account to buffer against temporary payment delays.
6. **Sponsor Sponsor Support Undertakings (SSU):** Binding corporate undertakings executed by the parent sponsor promising to:
    - Infuse equity to fund any construction cost overruns beyond the sanctioned project budget.
    - Fund any cash flow deficit required to maintain a minimum DSCR of $1.10x$.
    - Meet any shortfall in debt servicing caused by tax liabilities, termination penalties, or statutory delays prior to COD.

Copyright Notice: © 2026 www.loanbazaronline.com. All rights reserved.

5.2 The 7-Tier Trust and Retention Account (TRA) Escrow Waterfall

All cash inflows generated by the project SPV—including government construction grants, equity contributions, debt drawdowns, toll collections, tariff receipts, and insurance proceeds—are legally required to be deposited directly into a designated **Trust and Retention Account (TRA)** maintained with an Authorised Category-I Escrow Agent Bank.



Detailed TRA Waterfall Mechanics:

- **Tier 1 (Statutory Dues):** Payment of Goods and Services Tax (GST), TDS, property taxes, and statutory government royalties.
- **Tier 2 (Operating Outlays):** Payment of approved, LIE-certified O&M expenses, site operational payroll, utility bills, and annual property/loss-of-profit insurance premiums.
- **Tier 3 (Senior Interest):** Transfer of funds required to pay scheduled pre-tax interest obligations on senior project debt.
- **Tier 4 (Senior Principal):** Transfer of funds into the principal redemption account to meet scheduled semi-annual/quarterly debt amortization tranches.

Copyright Notice: © 2026 www.loanbazaronline.com. All rights reserved.

- **Tier 5 (DSRA Top-Up):** Replenishment of any drawdowns from the Debt Service Reserve Account back to the mandated 1-to-2 quarter buffer level.
- **Tier 6 (MMRA Reserves):** Provisioning into the Major Maintenance Reserve Account to fund periodic heavy overhaul expenses (e.g., highway resurfacing every 5 years or solar inverter replacements every 7 years).
- **Tier 7 (Promoter Surplus Distribution):** Any residual funds remaining in the TRA can be released to the parent promoter as dividends **only if all financial covenants are fully met** ($ ext{DSCR} 1.20x$, DSRA fully funded, zero default, and LIE compliance certificate issued).

5.3 Statutory Charge Registration & Substitution Rights

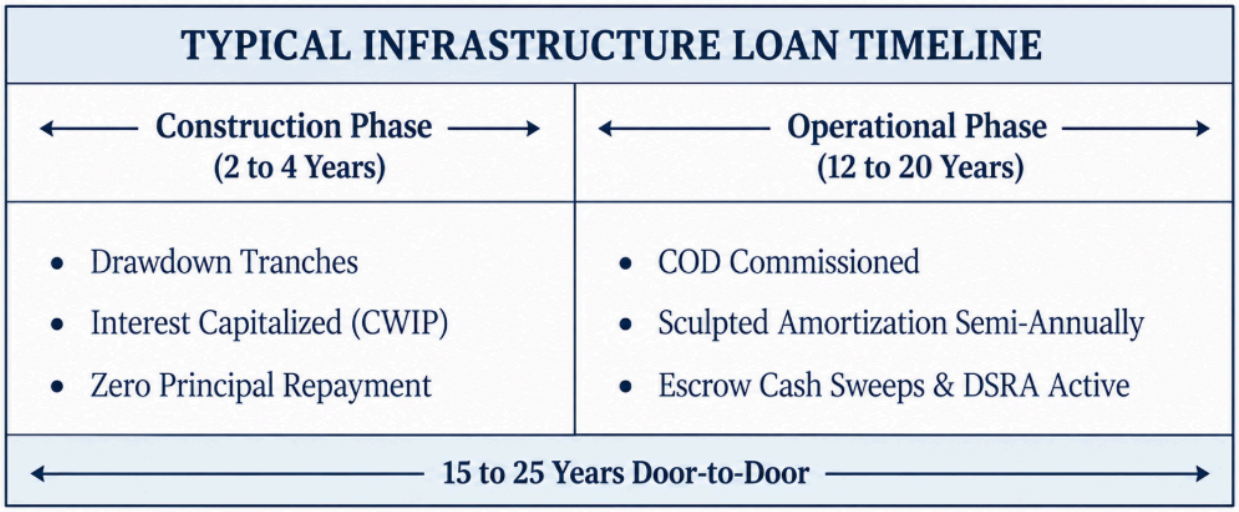
- **CERSAI & ROC Form CHG-1 Registrations:** Under Section 77 of the Companies Act, 2013, the security trustee must register the charge on the ROC portal via Form CHG-1 within 30 days of execution. Failure to register invalidates the charge against liquidators and competing creditors in insolvency. Charges must also be registered on the Central Registry of Securitisation Asset Reconstruction and Security Interest of India (CERSAI) portal.
- **Tripartite Substitution Rights Agreement:** Executed between the Concession Authority (e.g., NHAI/SECI), the Project SPV, and the Senior Lenders' Security Trustee. In the event of a material default by the promoter during construction or operation, senior lenders have the legally binding right to **step in and substitute the promoter** with a qualified third-party operator without terminating the underlying Concession Agreement/PPA, preserving the asset's economic continuity.

Copyright Notice: © 2026 www.loanbazaronline.com. All rights reserved.

Module 6: Indicative Tenure, Pricing Mechanics, Accounting & Tax Shields

Infrastructure Funding facilities feature distinctive commercial pricing parameters, extended door-to-door tenures, specialized corporate accounting standards, and powerful statutory tax shields under Indian income tax law.

6.1 Indicative Tenure, Moratorium & Sculpted Amortization



- **Door-to-Door Facility Tenure:** Spans **15 to 25 years**, structured to match the underlying 20-to-30 year Concession Agreement, PPA, or asset economic useful life.
- **Construction Moratorium Grace Period:** Spans **2 to 4 years** (matching the physical construction window up to COD plus an additional 3 to 6-month buffer). During the moratorium, the SPV makes zero principal repayments. Pre-COD interest costs are funded through sanctioned Interest During Construction (IDC) facility limits or capitalized into Capital Work-in-Progress (CWIP) under Ind AS 23.
- **Sculpted Amortization Profiles:** Principal repayment is not structured in flat equal monthly installments (EMIs). Instead, repayment is **sculpted** to match seasonal or ramping cash flows—starting with lower principal tranches in early operational years ($2\% - 4\% extperannum$) and stepping up in later years ($6\% - 8\% extperannum$) as traffic, tariff escalations, or annuity indexations increase project revenues.

6.2 Commercial Pricing Mechanics & Spread Sizing

The interest rate on Infrastructure Funding facilities is floating, benchmarked against bank repo-linked EBLR or Marginal Cost of Funds based Lending Rate (MCLR), plus a credit risk spread:

$$extAll - InBorrowingYield\left(K_d\right) = extFloatingBenchmarkRate(MCLR/EBLR/Repo) + extCreditSpread$$

Copyright Notice: © 2026 www.loanbazaronline.com. All rights reserved.

- **Construction Phase Pricing:** Ranging between **MCLR / EBLR** $+\ 2.00\%extto\ +\ 3.50\%extp.a.$ (effective $9.50\%extto11.50\%extp.a.$) reflecting construction completion, regulatory approval, and execution risks.
- **Post-DCCO De-risked Pricing:** Upon achieving COD and establishing 12 to 24 months of stable operational cash flows, the credit spread is contractually reset downward by **100 to 250 basis points**, lowering the operational borrowing rate to **EBLR / MCLR** $+\ 0.75\%extto\ +\ 1.75\%extp.a.$ (effective $8.00\%extto9.25\%extp.a.$).

6.3 Accounting Standards (Ind AS) & Service Concession Arrangements

1. Ind AS 23 (Borrowing Costs):

Requires all borrowing costs directly attributable to the acquisition, construction, or production of a qualifying infrastructure asset to be capitalized as part of the asset's Capital Work-in-Progress (CWIP) cost until COD. Upon achieving COD, capitalization ceases, and interest is expensed directly to the Profit & Loss Statement as a finance cost.

2. Ind AS 115 / Service Concession Arrangements (IFRIC 12 Compliance):

Applies to PPP projects (e.g., HAM, BOT Toll, Water Treatment) where a public grantor controls or regulates the infrastructure asset: * **Financial Asset Model (Applicable to HAM & BOT Annuity):** The SPV recognizes a financial asset (receivable) to the extent it has an unconditional contractual right to receive cash from the Concession Authority during the concession period. Revenue is recognized based on the fair value of construction services during construction, and as interest income + principal recovery during the annuity phase. * **Intangible Asset Model (Applicable to BOT Toll):** The SPV recognizes an intangible asset (Toll Collection Right) to the extent it receives a license to charge users of the public service. The intangible asset is amortized over the concession tenure using the traffic/revenue-proportionate amortization method.

6.4 Corporate Income Tax Shields & Statutory Depreciation

Interest paid on senior project debt is a fully tax-deductible operational expense under **Section 36(1)(iii) of the Income Tax Act, 1961**. This creates a substantial **Tax Shield**, reducing the SPV's net effective post-tax borrowing cost.

Pre-Tax vs. Post-Tax Cost of Debt Formula:

$$K_{d,extpost} = K_{d}imes\left(1\ -\ T_{extcorp}\right)$$

Copyright Notice: © 2026 www.loanbazaronline.com. All rights reserved.

Where K_d is the pre-tax interest rate, and $T_{extcorp}$ is the applicable effective corporate tax rate (including surcharge and cess). Under the corporate tax regime of **Section 115BAA** ($22\%extbasic + 10\%extsurcharge + 4\%extcess = 25.168\%$):

$$K_{d,extpost} = 9.50\%imes(1 - 0.25168) = 9.50\%imes0.74832 = 7.109\%extp.a.$$

Accelerated Depreciation Tax Shield (Section 32):

Under Section 32 of the Income Tax Act, 1961, specific infrastructure assets (e.g., renewable energy plants, water treatment equipment) qualify for accelerated tax depreciation rates of **40% per annum** on a Written Down Value (WDV) basis. This creates large non-cash tax losses in the early operational years, sheltering project cash flows from corporate income tax for the first 5 to 7 operational years.

Copyright Notice: © 2026 www.loanbazaronline.com. All rights reserved.

Module 7: Suitable Industries, Business Profiles & Sub-Sector Applications

Infrastructure Funding is suitable for capital-intensive, asset-heavy sub-sectors categorized under the RBI Harmonized Master List of Infrastructure:

PRIMARY SUB-SECTORS FOR INFRASTRUCTURE FUNDING		
1.	**TRANSPORT & LOGISTICS**	Expressways (HAM/BOT), Deep-Water Ports, Airports
2.	**ENERGY & UTILITIES**	Solar/Wind Hybrid Parks, Transmission Lines, Hydro
3.	**DIGITAL INFRASTRUCTURE**	Hyperscale Data Centers, Telecom Fiber Backbones
4.	**URBAN & SOCIAL INFRA**	Metro Rail, Water Treatment (STP), Smart Cities
5.	**LOGISTICS & STORAGE**	Multi-Modal Logistics Parks, Cold Chain Grids

7.1 Detailed Sub-Sector Applications & Structuring Highlights

1. **Transport & Logistics (Highways, Expressways, Bridges):**
   - *Ideal Structures:* NHAI Hybrid Annuity Model (HAM), BOT Toll, State Highway Concessions.
   - *Financing Focus:* $70{:}\,30 extto 75{:}\,25$ leverage; 15-year post-COD semi-annual annuity sculpting; 80%+ RoW land availability covenant.
2. **Energy & Power Utilities (Renewables, Transmission, Green Hydrogen):**
   - *Ideal Structures:* SECI / NTPC / DISCOM long-term 25-year PPA structures; Tariff-Based Competitive Bidding (TBCB) transmission lines.
   - *Financing Focus:* $75{:}\,25 extto 80{:}\,20$ leverage; 18-year post-COD door-to-door amortization; P-90 generation modeling; 40% accelerated depreciation tax shields under Section 32.
3. **Digital Infrastructure (Data Centers, Telecom Towers, Fiber Networks):**
   - *Ideal Structures:* Hyperscale Data Center Build-Own-Operate structures backed by long-term Master Service Agreements (MSAs) with global cloud hyperscalers (AWS, Azure, Google Cloud).

Copyright Notice: © 2026 www.loanbazaronline.com. All rights reserved.

- *Financing Focus:* $65:35 ext{ to } 70:30$ leverage; 10-to-15 year term debt; triple-net lease cash sweeps; strict uptime SLA monitoring.

4. **Urban Infrastructure, Water & Sanitation (Metro Rail, STPs, Desalination):**
   - *Ideal Structures:* Hybrid Annuity Model (HAM) City Sewage Treatment Plants (STPs) under the National Mission for Clean Ganga (NMCG) / Municipal Corporations.
   - *Financing Focus:* $70:30$ leverage; 12-to-15 year annuity amortization; municipal credit rating verification; tri-partite escrow accounts.

Copyright Notice: © 2026 www.loanbazaronline.com. All rights reserved.

Module 8: Key Advantages and Limitations of Infrastructure Funding

8.1 Strategic Advantages for Promoters & Lenders

- **Off-Balance Sheet Ring-Fencing for Sponsors:** Because debt is raised by a standalone SPV, the parent promoter's primary balance sheet remains unencumbered, preserving borrowing capacity for other core business expansions.
- **Extended Amortization Matching Asset Life:** Provides 15 to 25-year door-to-door tenures, ensuring that annual debt service obligations align with project cash generation, eliminating short-term refinancing risks.
- **High Leverage & Equity Efficiency:** Enables sponsors to raise $70\% extto 80\%$ senior debt against project cost, minimizing upfront equity dilution and maximizing Return on Equity (ROE).
- **Massive Post-Tax Cost Reduction via Tax Shields:** Full deductibility of pre-tax interest under Section 36(1)(iii) combined with Section 32 accelerated depreciation slashes net borrowing costs down to $7.00\% - 7.50\% extp.a.$
- **High Credit Quality Post-DCCO:** Operational infrastructure assets backed by government annuities or 25-year PPAs exhibit extremely low post-COD default rates, making them ideal assets for secondary market InvIT monetization.

8.2 Structural Limitations & Operational Constraints

- **High Front-Loaded Execution Risk:** Projects are vulnerable to multi-year construction delays, environmental litigation, land acquisition bottlenecks, and cost overruns during the pre-COD phase.
- **Complex Multi-Party Legal Framework:** Requires extensive legal, technical, and insurance documentation involving multiple stakeholders (Sponsors, Lenders, Concession Authority, EPC Contractor, O&M Contractor, Escrow Bank), leading to high upfront legal and structuring fees.
- **Strict Escrow TRA Cash Flow Restrictions:** The promoter has zero access to project revenues until Tiers 1 through 6 of the TRA waterfall (taxes, O&M, senior debt, DSRA, MMRA) are fully satisfied.
- **Concession & Tariff Re-negotiation Vulnerability:** Changes in government policy, state DISCOM tariff renegotiations, or judicial stays can disrupt SPV cash flows.

Copyright Notice: © 2026 www.loanbazaronline.com. All rights reserved.

Module 9: Comprehensive Real-World Case Study (Indian Rupee Models)

To illustrate the financial, structural, and tax dynamics of Infrastructure Funding, this module presents a detailed real-world comparative financial model for a major transportation infrastructure project in India.

9.1 Project Profile & Capital Outlay Architecture

Borrower Entity: *Mumbai & Konkan Highway Infrastructure SPV Ltd. (MKHISL)*
Sponsor: *Mumbai Infra Construction & Engineering Developers Ltd. (MICEDL)* (Net Worth: $ ext{₹1,200 Crores}$)
Project Scope: Development of a 120-km 4-lane Highway Expansion Project in Maharashtra under the National Highways Authority of India (NHAI) **Hybrid Annuity Model (HAM)**.
Concession Period: 17 Years (2 Years Construction Moratorium + 15 Years Operational Annuity Phase).

Project Cost & Capital Structuring ($ ext{₹}$ in Crores):

- **Total Project Outlay Budget (TPC):** $ ext{₹1,000.00 Crores}$
  - *NHAI Construction Grant (40% of TPC):* $ ext{₹400.00 Crores}$ (Paid in 5 milestone tranches of $ ext{₹80.00 Crores}$ each against $20\%, 40\%, 60\%, 80\%, 100\%$ physical construction progress).
  - *Net SPV Capital Outlay Required (60% of TPC):* $ ext{₹600.00 Crores}$

SPV Financing Stack ($ ext{₹600.00 Crore Outlay}$):

- **Sponsor Equity Contribution (25% of SPV Outlay / 15% of TPC):** $ ext{₹150.00 Crores}$
- **Senior SPV Infrastructure Term Loan (75% of SPV Outlay / 45% of TPC):** $ ext{₹450.00 Crores}$

MKHISL HAM PROJECT CAPITAL STACK (₹1,000 CRORES)		
NHAI Construction Grant (40%)	₹400.00 Crores	(5 Tranches of ₹80 Cr)
Senior SPV Project Loan (45%)	₹450.00 Crores	(Consortium Debt @ 9.50% p.a.)
Sponsor Equity Margin (15%)	₹150.00 Crores	(Promoter Cash Contribution)

Copyright Notice: © 2026 www.loanbazaronline.com. All rights reserved.

9.2 Comparative Financial Options Under Analysis

The sponsor evaluates two alternative capital structuring pathways to execute the project:

- **Pathway A (Sponsor Corporate Balance Sheet Term Loan):** The parent sponsor company borrows $ ext{₹450.00 Crores}$ directly on its own corporate balance sheet using a standard 5-year corporate term loan @ $12.50\% extp.a.$ pre-tax interest, with a 1-year principal grace period, requiring equal annual principal repayments of $ ext{₹112.50 Crores/year}$ starting Year 2.
- **Pathway B (Structured Non-Recourse SPV HAM Project Finance Consortium Loan):** The project SPV borrows $ ext{₹450.00 Crores}$ using a structured Non-Recourse SPV HAM Project Finance Facility @ $9.50\% extp.a.$ pre-tax interest during construction, stepping down to $8.25\% extp.a.$ post-COD, featuring a **3-year construction & stabilization moratorium** (principal repayment $ ext{₹0.00}$ during Y1–Y3; interest capitalized into CWIP under Ind AS 23), followed by a 14-year semi-annual annuity-sculpted amortization profile matching NHAI annuity receipts.

Copyright Notice: © 2026 www.loanbazaronline.com. All rights reserved.

9.3 Comparative Financial Modeling & Debt Servicing Schedule ($ ext{₹}$ in Crores)

YEAR-BY-YEAR FINANCIAL MODEL COMPARISON (₹ IN CRORES)					
METRIC / YEAR	**YEAR 1**	**YEAR 2**	**YEAR 3**	**YEAR 4**	**YEAR 5**
Project Status	Constr. Y1	Constr. Y2	COD Launch	Annuity Y1	Annuity Y2
Physical Construction Progress (%)	40.0%	100.0%	100.0%	100.0%	100.0%
NHAI Construction Grant Inflow	₹160.00	₹240.00	₹0.00	₹0.00	₹0.00
NHAI Bi-Annual Annuity Inflow	₹0.00	₹0.00	₹95.00	₹112.00	₹110.00
PATHWAY A: CORPORATE BALANCE SHEET LOAN (@ 12.50% p.a., 5-Year Amortization)					
Opening Corporate Debt Balance	₹450.00	₹450.00	₹337.50	₹225.00	₹112.50
Annual Pre-Tax Interest Expense	₹56.25	₹56.25	₹42.19	₹28.13	₹14.06
Scheduled Principal Amortization	₹0.00	₹112.50	₹112.50	₹112.50	₹112.50
Total Cash Debt Service Outflow	₹56.25	₹168.75	₹154.69	₹140.63	₹126.56
Project Operational EBITDA	₹0.00	₹0.00	₹95.00	₹112.00	₹110.00
Net Annual Cash Flow Surplus/Deficit	-₹56.25	-₹168.75	-₹59.69	-₹28.63	-₹16.56
Pathway A Project DSCR	0.00x	0.00x	0.61x	0.80x	0.87x
Financial Status	DEFICIT	DEFAULT!	INSOLVENT	INSOLVENT	DEFICIT
PATHWAY B: STRUCTURED NON-RECOURSE SPV HAM PROJECT LOAN (@ 9.50% / 8.25% p.a.)					
Opening SPV Debt Balance	₹450.00	₹450.00	₹450.00	₹425.00	₹395.00
Pre-Tax Interest (Capitalized Y1-Y2)	₹42.75	₹42.75	₹37.13	₹35.06	₹32.59
Scheduled Principal Amortization	₹0.00	₹0.00	₹25.00	₹30.00	₹35.00
Total Cash Debt Service Outflow	₹0.00	₹0.00	₹62.13	₹65.06	₹67.59
Project Operational EBITDA (CFADS)	₹0.00	₹0.00	₹92.00	₹108.00	₹106.00
Annual Post-Tax Interest Tax Shield	₹0.00	₹0.00	₹9.34	₹8.82	₹8.20
Net Post-Tax Debt Servicing Outflow	₹0.00	₹0.00	₹52.79	₹56.24	₹59.39
Net Retained Escrow Cash Surplus	₹0.00	₹0.00	+₹39.21	+₹51.76	+₹46.61
Pathway B Project DSCR	Moratorium	Moratorium	1.48x	1.66x	1.57x
Financial Status	SOLVENT	SOLVENT	PRIME COVER	PRIME COVER	PRIME COVER

Copyright Notice: © 2026 www.loanbazaronline.com. All rights reserved.

9.4 Deep Quantitative Financial Analysis & Strategic Valuation Findings

1. Cash Deficit & Default Avoidance (Year 1 to Year 3):

Under **Pathway A (Corporate Balance Sheet Debt)**, the parent sponsor faces a catastrophic cumulative cash deficit of $ ext{₹284.69 Crores}$ during Years 1 to 3 ($ ext{-₹56.25 Cr}$ in Y1, $ ext{-₹168.75 Cr}$ in Y2, and $ ext{-₹59.69 Cr}$ in Y3) because flat corporate debt repayments fall due during active construction when operational revenues are zero. This results in a failing DSCR of $0.00xexttto0.61x$, triggering immediate bank default and credit rating downgrade.

Under **Pathway B (Structured Non-Recourse SPV HAM Facility)**, the 3-year construction moratorium eliminates all cash principal and interest payments during construction (pre-COD interest of $ ext{₹42.75 Cr/yr}$ is capitalized into CWIP under Ind AS 23). Upon achieving COD in Year 3, NHAI annuity receipts ($ ext{₹92.00 Cr}$ CFADS) comfortably cover the sculpted debt service ($ ext{₹62.13 Cr}$), generating a prime operational **DSCR of 1.48x in Year 3, 1.66x in Year 4, and 1.57x in Year 5**, keeping the project fully solvent and compliant with all bank covenants!

2. Cost of Debt & Tax Shield Optimization:

Under Section 36(1)(iii) of the Income Tax Act, 1961, pre-tax interest paid on operational SPV debt generates a 25.168% corporate tax shield: * Pre-Tax Borrowing Rate (Post-COD): $8.25\%extp.a.$ * Effective Post-Tax Borrowing Rate:

$$K_{d,extpost} = 8.25\%imes(1 - 0.25168) = 8.25\%imes0.74832 = 6.173\%extp.a.$$

The interest tax shield saves the SPV **$ ext{₹9.34 Crores}$ in Year 3, $ ext{₹8.82 Crores}$ in Year 4, and $ ext{₹8.20 Crores}$ in Year 5**, directly enhancing net post-tax equity returns.

3. Total Net Strategic Financial Wealth Surplus Generated:

By choosing Pathway B (Structured Non-Recourse SPV Infrastructure Loan), the parent sponsor company achieves: * **Avoided Cash Deficits / Preserved Sponsor Liquidity (Y1–Y3):** $+ ext₹284.69Crores$ * **Post-Tax Interest Rate Differential Advantage (3.00% Spread Savings over 5 Yrs):** $+ ext₹41.52Crores$ * **Cumulative Retained Escrow Cash Surplus (Y3–Y5):** $+ ext₹137.58Crores$ * **ROCE Reinvestment Income Earned on Preserved Equity (@ 16.00% p.a. over Y1–Y5):** $+ ext₹91.80Crores$ * **Less Upfront Syndication & Structuring Fees (1.25%):** $- ext₹5.63Crores$ * **Net Strategic Financial Wealth Surplus Generated:** $+ ext₹549.96CRORES \quad (+ ext₹549,96,00,000.00)$

Copyright Notice: © 2026 www.loanbazaronline.com. All rights reserved.

Module 10: Regulatory Compliance, RBI Guidelines (2025/2026) & DCCO Framework

10.1 RBI Project Financing Directions (2025/2026 Prudential Framework)

To enhance credit resilience and prevent systematic non-performing asset formation in large-scale infrastructure projects, the Reserve Bank of India introduced updated prudential directions governing commercial bank and NBFC exposures to project finance:

1. Calibrated Standard Asset Provisioning Rules:

Lenders must maintain calibrated standard asset provisioning against outstanding infrastructure project debt based on the risk phase: * **Construction Phase Provisioning:** Standard asset provisioning is maintained at 5.0% **of total funded exposure** during active physical construction prior to COD. * **COD Step-Down Provisioning:** Upon successful physical completion and COD achievement, standard provisioning steps down to 2.5%. * **Operational Phase Provisioning:** Upon demonstrating two years of continuous operational cash flows with a minimum $ ext{DSCR} 1.25x$, provisioning drops to the standard baseline rate of 1.0%.

2. Permissible Date of Commercial Operation (DCCO) Extensions:

Recognizing that infrastructure projects face extraneous delays, RBI guidelines permit lenders to extend the original DCCO without automatically reclassifying the loan as a Non-Performing Asset (NPA), subject to strict conditions: * **Infrastructure Sector Extension Allowance:** DCCO can be extended by up to **2 years** for commercial infrastructure projects (and up to 4 years if delays are caused by court litigation or arbitration) without triggering NPA classification. * **Conditions for DCCO Deferment:** - The account must be a performing standard asset at the time of restructuring. - Independent technical consultant (LIE) certifies that the project is physically viable and delay is beyond promoter control. - Promoters infuse additional equity equal to at least 100% of the cost overruns resulting from the DCCO extension.

Copyright Notice: © 2026 www.loanbazaronline.com. All rights reserved.

Module 11: Professional Certification Examination & Detailed Rationales

This module contains 10 scenario-based examination questions designed to test comprehensive mastery of Infrastructure Funding, financial underwriting formulas, legal security structures, TRA escrow waterfalls, statutory tax shields, and RBI regulatory directives.

Question 1: Non-Recourse SPV Ring-Fencing & Bankruptcy Isolation

Scenario: A major Indian industrial conglomerate (*Sponsor Corp*) incorporates a standalone SPV (*Project Co Ltd.*) to construct a $ ext{₹2,000 Crore}$ deep-water port. Senior lenders syndicate $ ext{₹1,500 Crores}$ of non-recourse project debt. Two years into construction, Sponsor Corp enters insolvency under the Insolvency and Bankruptcy Code (IBC), 2016 due to heavy losses in an unrelated business vertical. Senior lenders of Project Co Ltd. are concerned about debt servicing. Which of the following statements correctly describes the legal status of Project Co Ltd. and its lenders?

A) Project Co Ltd. is automatically forced into liquidation under IBC alongside Sponsor Corp, and senior port debt is converted into sponsor equity.

B) As an independent bankruptcy-remote SPV with ring-fenced assets and an exclusive Trust and Retention Account (TRA), Project Co Ltd. remains legally insulated from Sponsor Corp's insolvency, allowing construction to continue provided equity commitments are met.

C) The liquidator of Sponsor Corp can freely withdraw funds from Project Co Ltd.'s escrow account to pay off Sponsor Corp's unsecured creditors.

D) The Concession Agreement is automatically terminated by government authorities upon sponsor insolvency without substitution rights.

Question 2: Financial Underwriting Ratios (DSCR Calculation)

Scenario: An operational highway SPV under the Hybrid Annuity Model (HAM) reports the following annual financial figures for Year 4: * EBITDA from Annuities & O&M Allowances: $ ext{₹120.00 Crores}$ * Interest Earned on Escrow TRA Deposits: $ ext{₹4.00 Crores}$ * Cash Corporate Taxes Paid: $ ext{₹8.00 Crores}$ * Mandatory Maintenance Capex: $ ext{₹6.00 Crores}$ * Scheduled Senior Debt Principal Amortization: $ ext{₹40.00 Crores}$ * Pre-Tax Senior Debt Interest Expense: $ ext{₹30.00 Crores}$

Copyright Notice: © 2026 www.loanbazaronline.com. All rights reserved.

What is the SPV's Debt Service Coverage Ratio (DSCR) for Year 4?

A) $1.15x$

B) $1.35x$

C) $1.57x$

D) $1.71x$

Question 3: Loan Life Coverage Ratio (LLCR) Mechanics

Scenario: An infrastructure credit analyst is evaluating a $ ext{₹1,000 Crore}$ solar power SPV. The outstanding senior debt balance is $ ext{₹600 Crores}$. The net present value (NPV) of future Cash Flows Available for Debt Service (CFADS) discounted at the weighted average cost of debt over the remaining loan amortization period (12 years) is $ ext{₹900 Crores}$. The NPV of CFADS calculated over the total 25-year concession life is $ ext{₹1,150 Crores}$. What is the Loan Life Coverage Ratio (LLCR) of the SPV?

A) $1.25x$

B) $1.50x$

C) $1.83x$

D) $1.92x$

Question 4: Trust and Retention Account (TRA) Escrow Waterfall Priority

Scenario: An operational port SPV receives $ ext{₹50.00 Crores}$ in user charges in its bank-controlled TRA Escrow Account during Q2. The following payment requests are submitted simultaneously: 1. Dividend distribution to promoter sponsor: $ ext{₹10.00 Crores}$ 2. Statutory GST and port authority royalties: $ ext{₹6.00 Crores}$ 3. Scheduled senior debt interest and principal: $ ext{₹20.00 Crores}$ 4. Essential site operating expenses & insurance premiums: $ ext{₹8.00 Crores}$

According to standard 7-tier TRA waterfall covenants, in what exact order must the escrow agent bank release these payments?

Copyright Notice: © 2026 www.loanbazaronline.com. All rights reserved.

A) Item 2 (Statutory Dues) $ ightarrow$ Item 4 (Essential O&M) $ ightarrow$ Item 3 (Senior Debt) $ ightarrow$ Item 1 (Promoter Dividends)

B) Item 1 (Promoter Dividends) $ ightarrow$ Item 3 (Senior Debt) $ ightarrow$ Item 2 (Statutory Dues) $ ightarrow$ Item 4 (Essential O&M)

C) Item 3 (Senior Debt) $ ightarrow$ Item 2 (Statutory Dues) $ ightarrow$ Item 4 (Essential O&M) $ ightarrow$ Item 1 (Promoter Dividends)

D) Item 4 (Essential O&M) $ ightarrow$ Item 1 (Promoter Dividends) $ ightarrow$ Item 3 (Senior Debt) $ ightarrow$ Item 2 (Statutory Dues)

Question 5: Section 36(1)(iii) Interest Tax Shield Computation

Scenario: A renewable energy SPV holds an outstanding senior debt balance of $ ext{₹800.00 Crores}$ carrying a floating pre-tax interest rate of $9.00\% extp.a.$ The SPV operates under the corporate tax regime of Section 115BAA with an effective corporate tax rate of 25.168%. What is the annual monetary tax shield value saved by the SPV, and what is its effective post-tax cost of debt ($K_{d,extpost}$)?

A) Pre-Tax Interest: $ ext{₹72.00 Cr}$; Tax Shield: $ ext{₹18.12 Cr}$;
$K_{d,extpost} = 6.735\% extp.a.$

B) Pre-Tax Interest: $ ext{₹80.00 Cr}$; Tax Shield: $ ext{₹20.13 Cr}$;
$K_{d,extpost} = 7.200\% extp.a.$

C) Pre-Tax Interest: $ ext{₹72.00 Cr}$; Tax Shield: $ ext{₹10.00 Cr}$;
$K_{d,extpost} = 8.100\% extp.a.$

D) Pre-Tax Interest: $ ext{₹64.00 Cr}$; Tax Shield: $ ext{₹16.10 Cr}$;
$K_{d,extpost} = 5.950\% extp.a.$

Question 6: Statutory Security Registration & CERSAI / Form CHG-1 Rules

Scenario: A consortium of banks sanctions a $ ext{₹500 Crore}$ project loan to a telecom tower SPV. The security trustee executes the Mortgage Deed and Hypothecation Agreement on

Copyright Notice: © 2026 www.loanbazaronline.com. All rights reserved.

January 10. Due to an administrative oversight, the security trustee fails to file ROC Form CHG-1 or register the charge on CERSAI within 30 days. On March 15, the SPV enters liquidation under IBC. What is the legal implication of this filing failure?

A) The security charge remains fully valid, and senior lenders retain first-priority mortgage rights over liquidators.

B) Under Section 77 of the Companies Act, 2013, the unregistered charge is void against the liquidator and competing creditors, reducing the senior lenders to unsecured financial creditor status for asset distribution.

C) The Registrar of Companies automatically registers the charge and issues a fine of $ ext{₹1,000}$ without altering security priority.

D) The mortgage is automatically transferred to the parent sponsor company.

Question 7: RBI Project Financing Directions (2025/2026 Provisioning Rules)

Scenario: A commercial bank participates in a $ ext{₹1,200 Crore}$ syndicated project loan for a greenfield metro rail SPV. The construction phase is scheduled for 3 years. According to the updated RBI Project Financing Directions (2025/2026), what percentage of standard asset provisioning must the bank maintain during active construction, and what rate does it step down to upon achieving COD?

A) 1.0% during construction, stepping down to 0.4% at COD.

B) 5.0% during construction, stepping down to 2.5% at COD.

C) 10.0% during construction, stepping down to 5.0% at COD.

D) 2.0% during construction, stepping down to 1.0% at COD.

Question 8: Hybrid Annuity Model (HAM) Payment Structure

Scenario: An EPC developer is executing a $ ext{₹500 Crore}$ NHAI HAM road project. During the 2-year construction phase, how much cash grant does NHAI provide directly to the SPV, and how is the residual 60% balance reimbursed post-COD?

Copyright Notice: © 2026 www.loanbazaronline.com. All rights reserved.

A) NHAI provides a 100% upfront cash grant during construction; zero payments are made post-COD.

B) NHAI provides a 40% cash grant in 5 milestone tranches ($ ext{₹200 Cr}$ total) during construction; the remaining 60% is reimbursed via semi-annual inflation-indexed annuities over 15 years with interest on residual balances.

C) NHAI provides zero construction grants; the developer collects tolls directly from commercial traffic.

D) NHAI provides a 20% grant; the remaining 80% is converted into equity shares held by NHAI.

Question 9: Ind AS 23 Borrowing Cost Capitalization

Scenario: An express tower SPV incurs $ ext{₹40.00 Crores}$ in senior debt interest during its 2-year construction phase prior to achieving COD. How must this interest expense be treated under Ind AS 23 and presented in the financial statements?

A) Fully expensed to the Profit & Loss statement in Year 1 as an immediate operational finance loss.

B) Capitalized as part of Capital Work-in-Progress (CWIP) on the Balance Sheet during construction, and amortized via depreciation post-COD over the asset's useful life.

C) Subtracted directly from promoter equity share capital without appearing on the Balance Sheet.

D) Converted into a government subsidy receivable.

Question 10: RBI Permissible DCCO Extension Rules

Scenario: A 250 MW wind power SPV experiences an 18-month construction delay beyond its original Date of Commercial Operation (DCCO) due to a prolonged judicial injunction on land acquisition in the High Court. The loan is currently standard and performing. Can the lender group extend the DCCO without reclassifying the loan as an NPA under RBI IRACP rules?

Copyright Notice: © 2026 www.loanbazaronline.com. All rights reserved.

A) No, any delay beyond 30 days automatically triggers 100% NPA classification.

B) Yes, RBI guidelines permit DCCO extensions up to 2 years for commercial/infrastructure projects (and up to 4 years for court litigation delays) without NPA classification, provided the account is standard, the delay is beyond promoter control, and promoters fund cost overruns.

C) Yes, but only if the entire senior debt is converted into equity.

D) No, the Concession Agreement must be cancelled and re-auctioned.

Detailed Answer Key & Step-by-Step Educational Rationales

Answer 1: B

Rationale: Under the Special Purpose Vehicle (SPV) architecture, Project Co Ltd. is a legally independent, ring-fenced, bankruptcy-remote entity. Its assets, Concession Agreement, and cash inflows in the Trust and Retention Account (TRA) are legally insulated from the parent sponsor. The insolvency of Sponsor Corp under IBC does not force Project Co Ltd. into liquidation, nor can the sponsor's liquidator access the SPV's escrow cash flows. Provided equity call commitments are fulfilled (by sponsor or standby LOCs), senior project lenders remain fully protected.

Answer 2: C

Rationale: 1. First, calculate Cash Flow Available for Debt Service (CFADS):

$extCFADS = extEBITDA(ext₹120.00Cr) + extInterestIncome(ext₹4.00Cr) - extTaxes(ext₹8.00Cr)$

2. Second, calculate Total Senior Debt Service:

$extTotalDebtService = extPrincipalAmortization(ext₹40.00Cr) + extPre - TaxInterest(ext₹30.00$

3. Third, compute DSCR:

$$ ext{DSCR} = rac{ ext{CFADS}}{ ext{Debt Service}} = rac{ ext{₹110.00 Cr}}{ ext{₹70.00 Cr}} = \mathbf{1.5714x \quad (pprox 1.57x)}$$

Answer 3: B

Rationale: The Loan Life Coverage Ratio (LLCR) is defined as the NPV of future CFADS discounted at the weighted average cost of debt over the remaining loan life divided by the outstanding senior debt balance:

Copyright Notice: © 2026 www.loanbazaronline.com. All rights reserved.

$extLLCR = racextNPVofCFADSoverLoanAmortizationPeriodextOutstandingDebtBalance = racext$

(Note: The $ ext{₹1,150 Crore}$ NPV over concession life represents the PLCR of $1.92x$*).*

Answer 4: A

Rationale: Under standard 7-tier Trust and Retention Account (TRA) escrow covenants: 1. Tier 1: Statutory Taxes & Royalties ($ ext{Item 2}$) must be satisfied first. 2. Tier 2: Essential O&M Expenses & Insurance ($ ext{Item 4}$) are paid next to ensure physical asset operation. 3. Tier 3 & Tier 4: Senior Debt Interest & Principal ($ ext{Item 3} $)$areservicedthird.4.Tier7:Dividenddistributionstopromotersponsors($ ext{Item 1}$) are strictly subordinated at the bottom of the waterfall and released only after senior debt obligations and reserves are fully satisfied. Therefore, the correct sequence is **Item 2 $ ightarrow$ Item 4 $ ightarrow$ Item 3 $ ightarrow$ Item 1**.

Answer 5: A

Rationale: 1. Compute Pre-Tax Annual Interest Expense:

$$extPre - TaxInterest = ext₹800.00Croresimes9.00\% = ₹72.00extCrores$$

2. Compute Corporate Tax Shield Savings under Section 36(1)(iii) @ 25.168%:

$$extAnnualTaxShield = ext₹72.00Croresimes25.168\% = ₹18.12extCrores$$

3. Compute Effective Post-Tax Cost of Debt ($K_{d,extpost}$):

$$K_{d,extpost} = 9.00\%imes(1 - 0.25168) = 9.00\%imes0.74832 = 6.735\%extp.a.$$

Answer 6: B

Rationale: Under Section 77 of the Companies Act, 2013, every company creating a charge on its property or assets must register the charge with the Registrar of Companies (ROC) via Form CHG-1 within 30 days. If a charge is not registered, Section 77(3) explicitly dictates that the charge **shall be void against the liquidator and any creditor of the company**. Consequently, upon SPV liquidation under IBC, the lenders lose their mortgage priority and are relegated to unsecured financial creditor status.

Answer 7: B

Rationale: Under the Reserve Bank of India (Project Financing) Directions 2025/2026, regulated commercial lenders must maintain a calibrated standard asset provisioning of **5.0% of total funded exposure during the active construction phase**. Upon achieving successful physical completion and COD, standard provisioning steps down to **2.5%**, and further decreases to **1.0%** after two years of operational stability ($ ext{DSCR} 1.25x$).

Copyright Notice: © 2026 www.loanbazaronline.com. All rights reserved.

Answer 8: B

Rationale: Under the NHAI Hybrid Annuity Model (HAM), the government Concession Authority provides a **40% construction grant** disbursed in 5 equal milestone payments of 8% each during the construction phase based on certified physical progress. The residual 60% balance is funded upfront by the SPV via debt/equity, and is subsequently reimbursed post-COD by NHAI through 30 semi-annual inflation-indexed annuities over 15 years, along with interest on the residual balance (benchmarked to RBI Bank Rate + spread).

Answer 9: B

Rationale: Under Ind AS 23 (Borrowing Costs), interest costs directly attributable to the construction of a qualifying asset must be capitalized into Capital Work-in-Progress (CWIP) on the Balance Sheet throughout the construction phase prior to COD. Once the asset is commissioned (DCCO), capitalization ceases, CWIP is reclassified into Fixed Assets, and capitalized interest is amortized over the asset's useful life through depreciation.

Answer 10: B

Rationale: Under RBI IRACP and Project Finance directions, commercial lenders are permitted to grant DCCO extensions of **up to 2 years for infrastructure projects** (and up to 4 years if delays are attributable to court litigation or land acquisition injunctions) without treating the modification as a restructuring or classifying the account as an NPA. This requires that the project remains physically viable, the account is standard at the time of extension, and promoters infuse additional equity to cover cost overruns.

Copyright Notice: © 2026 www.loanbazaronline.com. All rights reserved.